

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA0508

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Concept Mapping (Medium)

Assessment Tool Weightage: 35%

Dr. Ayyappan G

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	5
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	5
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	5
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	5
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	5
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	5

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	5
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	5
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	5
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	5

Code of Conduct:

I Hemalakshmi H/192571049 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Concept Map: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship

Database Management System (DBMS)

A DBMS manages data using **data models**. A **schema** describes the overall structure of the database, while an **instance** represents the data stored at a particular moment. The **ER model** is one type of data model, represented using **ER diagrams**. ER diagrams contain **entities**, their **attributes**, and **relationships** between entities.

Structured points:

- **DBMS** → software to store, manage, and retrieve data
 - **Data Model** → defines how data is structured (ER model is one type)
 - **Schema** → overall design/blueprint of the database
 - **Instance** → actual data at a specific time
 - **ER Diagram** → visual representation of the ER data model
 - **Entity** → real-world object
 - **Attribute** → properties of an entity
 - **Relationship** → association between entities
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2. Concept Map: ANSI/SPARC Three-Level Architecture

ANSI/SPARC Architecture

The ANSI/SPARC architecture has **three levels**:

- **External Level** → user views
- **Conceptual Level** → logical structure
- **Internal Level** → physical storage

Data independence:

- **Logical Data Independence** → changes in conceptual level do not affect external views
- **Physical Data Independence** → changes in internal level do not affect conceptual level

Structured points:

- External ↔ Conceptual ↔ Internal
 - External → user-specific views
 - Conceptual → global logical schema
 - Internal → physical file organization
 - Logical data independence → conceptual ↔ external separation
 - Physical data independence → internal ↔ conceptual separation
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3. Concept Map: File-Based System vs Database System

File-Based System

File-based systems store data in separate files, causing problems like redundancy, inconsistency, and lack of security. DBMS solves these by providing centralized control, data integrity, and efficient access.

Problems in file-based systems:

- Data redundancy

- Data inconsistency
- No concurrency control
- No security
- Difficult data access
- No backup/recovery

DBMS solutions:

- Centralized database
 - Integrity constraints
 - Concurrency management
 - Security & authorization
 - Query processing
 - Backup & recovery
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4. Concept Map: ER Diagram Components

ER Diagram

ER diagrams contain different types of entities, attributes, relationships, and constraints.

Components:

- **Entity** → strong entity with its own key
 - **Weak Entity** → depends on strong entity, has partial key
 - **Attribute Types:**
 - Simple / Composite
 - Single-valued / Multi-valued
 - Derived
 - **Relationship Types:**
 - Unary, Binary, Ternary
 - Degree of relationship
 - **Participation Constraints:**
 - Total participation
 - Partial participation
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5. Concept Map: Hierarchical, Network, and Relational Data Models

Hierarchical Model

- Tree structure
- Parent–child relationship
- Fast navigation
- Rigid structure

Network Model

- Graph structure
- Many-to-many relationships
- Flexible links
- Complex to design

Relational Model

- Tables (relations)

- Uses keys and constraints
- High flexibility
- SQL support
- Most widely used

Comparison points:

- Structure: Tree vs Graph vs Tables
 - Flexibility: Low vs Medium vs High
 - Relationships: 1:N vs M:N vs Any
 - Ease of use: Hard vs Moderate vs Easy
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6. Concept Map: Enhanced Entity-Relationship (EER) Model

EER Model

EER extends ER by adding advanced concepts.

Generalization:

- Combining multiple entities into a higher-level entity
- Bottom → Up approach

Specialization:

- Dividing an entity into sub-entities
- Top → Down approach

Aggregation:

- Treating a relationship as an entity
 - Used for complex relationships
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7. Concept Map: DBMS Software Architecture

Flow: User → Query Processor → Execution Engine → Storage Manager → File System → Data → Back to User

Components:

- **Query Processor**
 - Parser
 - Optimizer
 - Executor
 - **Storage Manager**
 - Buffer manager
 - File manager
 - Index manager
 - **Transaction Manager**
 - Concurrency control
 - Recovery
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8. Concept Map: Data Models and Real-World Applications

Hierarchical Model:

- File systems
- XML data

- Organization charts

Network Model:

- Telecom networks
- Airline reservation systems

Relational Model:

- Banking
- E-commerce
- College databases

ER/EER Models:

- Conceptual design for any application
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9. Concept Map: Schema, Instance, and Data Independence**Schema:**

- Logical design
- Stable
- Three types: physical, logical, external

Instance:

- Actual data at a moment
- Changes frequently

Data Independence:

- **Physical** → internal changes don't affect logical schema
 - **Logical** → logical changes don't affect external views
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10. Concept Map: Roles and Responsibilities in a DBMS Environment**DBA (Database Administrator):**

- Security
- Backup
- Performance tuning
- Schema definition

End User:

- Access data
- Run queries
- Use applications

Application Developer:

- Write programs
- Design forms
- Implement business logic